

Reducing Delivery Lead Time with Machine Learning: Lead-Time Prediction, Shipment Simulation and Payback-Based Factory Reallocation for Nassau Candy Distributor

Amara Suchitra

Department of Artificial Intelligence and Machine Learning

Sai Vidya Institute of Technology

Bengaluru, India

amarasuchitra@gmail.com

Abstract—Moving 42 product–region routes to a different factory would make them both faster and cheaper, for an estimated profit uplift of \$28,507 on the historical order volume. Nassau Candy Distributor ships 15 products from five factories to four US regions, and each product is tied to one factory regardless of where the customer is. Using 10,194 orders from 2025, we estimate shipping distance, train three regression models to predict lead time, and simulate every product, region, factory and ship-mode combination (1,060 scenarios). Linear Regression predicts best ($R^2 = 0.972$, $MAE = 0.24$ days). Candidate moves are ranked by lead-time reduction, profit impact and data confidence. Because changing a route’s factory is not free, a payback rule then tests each move against a one-time switching cost: at \$1,500 per route and a 12-month payback limit, 10 of 53 routes are worth moving, saving \$24,599 a year and paying back in 4–12 months. K-means clustering independently groups the routes into four types, and all eight long-haul, high-gain routes it finds pass the payback test. The results are delivered as a public decision-support web application with an animated shipment simulation, deployed on Streamlit Community Cloud.

Index Terms—supply chain, factory allocation, lead-time prediction, regression, simulation, k-means clustering, payback analysis, decision support.

I. INTRODUCTION

Factory choice is the largest lever on delivery time that a distributor controls, and where each product is produced and shipped from is a core supply-chain design decision [1, 2]. Nassau Candy assigns each product to a single factory, so an order from a Pacific customer may cross the country even when another factory able to make the same product is much closer.

This work asks one question: for each product and destination region, which factory should ship it? The objectives are to (1) estimate shipping distance and lead time for every order, (2) train and compare models that predict lead time, (3) simulate every alternative factory for every product–region pair and ship mode, (4) rank reallocations by speed gain, profit impact

TABLE I
DATASET COMPOSITION (ORDER LINES)

Dimension	Values
Regions	Pacific 3,253; Atlantic 2,986; Interior 2,335; Gulf 1,620
Ship modes	Standard 6,120; Second 1,979; First 1,548; Same Day 547
Factories	Lot’s O’ Nuts, Wicked Choccy’s, Sugar Shack, Secret Factory, The Other Factory
Products	15, each mapped to one current factory
Fields	Order, customer, product and financial fields (Sales, Units, Gross Profit, Cost)

and confidence, (5) keep only moves whose savings repay the one-time cost of switching, (6) group routes into types a manager can act on, and (7) deliver the findings as an interactive application usable without code.

Classic facility-location research optimises such assignments under cost and distance [3, 4], while recent work shows that predictive analytics and machine learning improve logistics decisions [5–7]. This work combines both: a learned lead-time model feeding a scenario simulation.

The contributions are a reproducible pipeline; a ranked list of 212 candidate moves; a switching-cost payback rule that turns them into three decisions (Reallocate, Pilot first, Keep current); a route clustering that explains where the gains lie; and a deployed application that shows each move on a map and exports an action plan.

II. DATA

The Nassau Candy Distributor dataset contains 10,194 order lines placed between 1 January and 31 December 2025, with 18 fields per line (Table I). Chocolate dominates volume (9,844 lines), followed by Other (310) and Sugar (40). The dataset has no factory, distance or shipping-cost field; the product-to-factory mapping and factory coordinates come from the project brief, and customers are located at state centroids.

III. METHODOLOGY

The pipeline is deterministic (random seed 42), so every figure here can be reproduced with one command.

A. Preparation and Distance

Dates were parsed day-first. The raw gap between order and ship dates contained whole-year offsets, removed by subtracting the nearest multiple of 365.25 days. Shipping distance d is the great-circle (haversine) distance [8] in miles from the factory to the customer’s state centroid.

B. Lead-Time Target

After correction, recorded lead time barely varied (median 177.5 days, standard deviation 5.3) and was uncorrelated with distance ($r = -0.03$), so the dates do not measure shipping performance. We therefore construct a transparent modelled lead time from the two real levers, ship mode m and distance d , plus a division handling allowance $h(v)$ and noise:

$$LT = b(m) + \frac{d}{500} + h(v) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 0.3^2), \quad (1)$$

floored at 0.3 days, where $b(m)$ is 0.5, 1.5, 2.5 and 4.0 days for Same Day, First, Second and Standard Class, and $h(v)$ is 0.4, 0.3 and 0.2 days for Chocolate, Other and Sugar. Shipping cost uses a documented proxy of \$0.001 per mile per unit.

C. Models

Three regressors from scikit-learn [9] predict LT from distance, units and sales (standardised) and product, factory, region, ship mode and division (one-hot encoded): Linear Regression, Random Forest [10] and Gradient Boosting [11]. Data was split 80/20 into training and test sets and models were compared on RMSE, MAE and R^2 [12].

D. Simulation and Scoring

Following standard what-if simulation practice [13], the best model predicts lead time for all 1,060 combinations of 53 observed product–region pairs, 5 factories and 4 ship modes. Each alternative is compared with the current factory. Confidence grows with the number of historical orders n :

$$c = \min\left(1, \frac{\ln(1+n)}{\ln(1+500)}\right), \quad (2)$$

and moves are ranked by a composite score

$$S = 0.6 \max(0, \Delta LT\%) + 0.2 \operatorname{sgn}(\Delta P) \min(20|\Delta P|, 20) + 20c, \quad (3)$$

where $\Delta LT\%$ is the lead-time reduction and ΔP the profit impact per order. A route is *win-win* when its best alternative is both faster and cheaper.

E. Switching Cost and Payback

Moving a route to another factory has a one-time cost (setting up production, moving stock) that the dataset does not record.

TABLE II
LEAD-TIME MODEL PERFORMANCE ON THE TEST SET

Model	RMSE (days)	MAE (days)	R^2
Linear Regression	0.304	0.241	0.972
Gradient Boosting	0.311	0.245	0.970
Random Forest	0.328	0.259	0.967

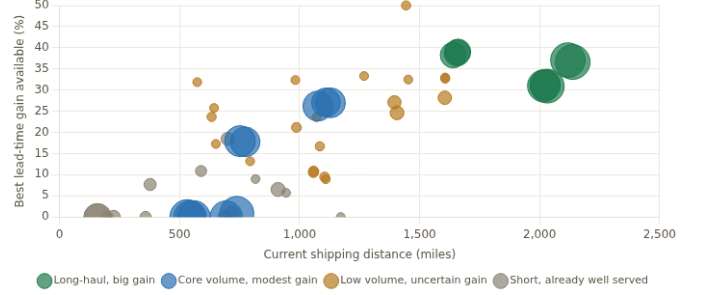


Fig. 1. Route groups: current shipping distance against the best lead-time gain available; bubble size shows 2025 orders.

We therefore treat it as an explicit business assumption K that the user can change. For each route’s best move, the yearly saving is $S = \Delta P \times n$, with n the route’s 2025 orders, and the payback period in months is

$$T = \frac{12K}{S}. \quad (4)$$

Each move receives one of three decisions. *Reallocate*: faster, $S > 0$, $T \leq L$ and confidence $c \geq 0.3$. *Pilot first*: worthwhile but $c < 0.3$ or $L < T \leq 2L$. *Keep current*: not faster, $S \leq 0$, fewer than $n_{\min}$ orders a year, or $T > 2L$. Defaults are $K = \$1,500$, $L = 12$ months and $n_{\min} = 20$ orders.

F. Route Clustering

To describe where the gains lie, the 53 routes (current factory, Standard Class) are clustered with k-means on three standardised features: shipping distance, $\ln(1+n)$ for order volume, and the best lead-time gain available. Initialisation is deterministic (the busiest route, then repeatedly the farthest point), so the groups are reproducible. Silhouette scores for $k = 2-6$ were 0.41, 0.42, 0.34, 0.38 and 0.37; we use $k = 4$, which scores close to the best while separating high-volume core routes from long-haul routes, a distinction the payback analysis needs.

IV. RESULTS

A. Model Comparison

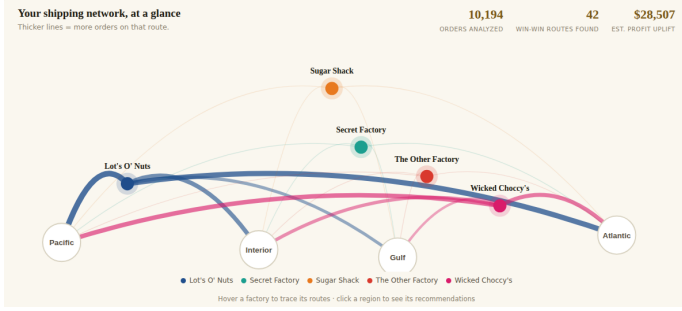
All three models explain about 97% of test-set variance (Table II). Linear Regression is best because the target is additive in distance and ship mode, which a linear model captures exactly; the ensembles add no accuracy and are harder to explain. It is used for every simulation.

B. Reallocation Recommendations

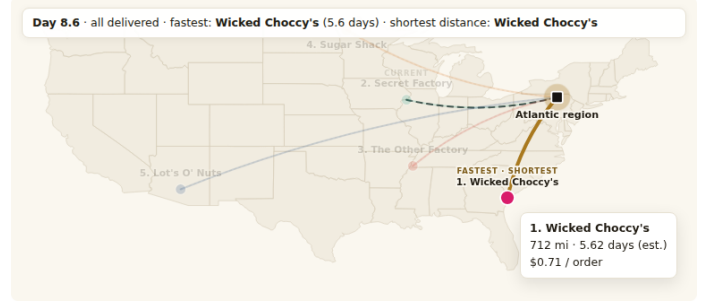
The simulation produced 212 candidate moves, 111 of them actionable. The five highest-scoring moves shorten lead time by 36–39% (Table III). The pattern is geographic: Pacific orders are best served from Lot’s O’ Nuts (Arizona) and Gulf orders from Wicked Choccy’s (Georgia). Taking the best alternative for each product–region pair, 42 routes are both faster and cheaper; applied to 2025 volume they yield an estimated profit uplift of \$28,507.

TABLE III
TOP FIVE REALLOCATION RECOMMENDATIONS

Product	Region	Current factory	Recommended factory	LT cut	$\Delta P/\text{order}$	Orders	Conf.
Wonka Bar – Milk Chocolate	Pacific	Wicked Choccy's	Lot's O' Nuts	37.1%	+\$6.48	666	1.00
Wonka Bar – Triple Dazzle Caramel	Pacific	Wicked Choccy's	Lot's O' Nuts	36.6%	+\$6.10	676	1.00
Wonka Bar – Scrumdiddlyumptious	Gulf	Lot's O' Nuts	Wicked Choccy's	38.8%	+\$5.81	310	0.92
Wonka Bar – Fudge Mallows	Gulf	Lot's O' Nuts	Wicked Choccy's	39.0%	+\$5.60	291	0.91
Wonka Bar – Nutty Crunch Surprise	Gulf	Lot's O' Nuts	Wicked Choccy's	38.3%	+\$5.41	296	0.92



(a) Current shipping network; line weight shows order volume per factory–region route.



(b) Shipment simulation, Everlasting Gobstopper to the Atlantic region (Standard Class): Wicked Choccy's arrives first (5.6 days); the dashed line is the current route.

Fig. 2. The decision-support application: (a) the network overview and (b) the animated shipment simulation.

TABLE IV
SENSITIVITY TO THE SWITCHING COST (12-MONTH PAYBACK)

Switching cost	Reallocate	Pilot first	Keep	Yearly saving
\$1,000	12	1	40	\$27,059
\$1,500	10	3	40	\$24,599
\$2,000	5	7	41	\$16,516
\$3,000	3	7	43	\$11,472

C. Risk Flags

One recommendation shortens lead time but raises estimated shipping cost per order. Eighty recommendations rest on few historical orders ($c < 0.3$) and should be piloted before a full move.

D. Decisions After Switching Cost

Accounting for the cost of switching changes the picture sharply. Of the 42 win-win routes, only 10 repay a \$1,500 switch within 12 months; 3 more should be piloted and 40 routes keep their current factory. The 10 moves save \$24,599 a year, or \$9,599 net in the first year, with payback between 4.2 and 12.0 months. The result is sensitive to the assumed cost (Table IV), which is why the application lets managers set it.

E. Route Groups

The four clusters (Table V, Fig. 1) give each decision a reason. All 8 *long-haul, big gain* routes are shipped about 1,900 miles today and receive *Reallocate*. *Core volume* routes carry half of all orders but gain less, so only 2 of 10 pass and 3 are piloted. The 21 *low-volume* routes show large percentage gains on only 129 orders in total, too few to repay a switch, and the 14 *short* routes are already served from a nearby factory. That two independent methods, one geometric and one financial, single out the same routes strengthens the recommendation.

TABLE V
ROUTE GROUPS FROM K-MEANS CLUSTERING

Group	Routes	Orders	Avg mi	Gain	Decisions
Long-haul, big gain	8	3,900	1,911	35.4%	8 R
Core volume	10	5,281	791	11.7%	2 R, 3 P, 5 K
Low volume	21	129	1,121	24.1%	21 K
Short, well served	14	884	572	4.2%	14 K

R = Reallocate, P = Pilot first, K = Keep current; gain = average best lead-time gain.

V. DECISION-SUPPORT APPLICATION

The findings are delivered as a public web application¹ so a manager can check any product and region without code. It opens with the headline numbers (10,194 orders, 42 win-win routes, \$28,507 uplift) and a map of the current network (Fig. 2a), followed by four sections: a *Factory Simulator* (recommended factory, predicted lead time for all factories and a ranking), *Compare* (current versus any alternative across regions), *Recommendations* (top opportunities, the payback decisions, the route groups and the ranked list) and a *Risk & Impact Panel*. In the payback card, sliders set the switching cost, payback limit and minimum volume; the decision counts, savings and table update at once, and the list can be downloaded as a CSV action plan.

In the shipment simulation (Fig. 2b), all five factories dispatch the same order on day 0 and each shipment travels at its predicted lead time on a map of the United States. A day counter reports arrivals in order; the first to arrive is the recommended factory and its route turns gold. Users can replay the run, scrub through days, and hover a factory for its distance, lead time and cost; the run replays whenever the product, region or ship mode changes.

¹<https://candy-factory-allocation.streamlit.app>

The pipeline writes one data bundle read by the page. A FastAPI backend serves the page and a JSON API; the same page is hosted on Streamlit Community Cloud from the GitHub repository² and redeploys on every push. An automated suite of 17 tests checks the API and page on each change.

VI. LIMITATIONS AND FUTURE WORK

Lead time is modelled, not measured: the transit speed, ship-mode base days and handling allowances are assumptions that real carrier data would replace. Shipping cost is a per-mile proxy that ignores freight rates and surcharges. Customers are placed at state centroids, understating distance variation within large states. The simulation assumes any factory has capacity to make any product, and sparse data for Sugar Shack and The Other Factory explains the 80 low-confidence moves. The switching cost is an assumption rather than data, and per-order effects such as cancellations and returns, as well as inventory positioning (stocking high-demand products near a region instead of moving production), are outside the scope of this dataset. Future work will add carrier transit times and freight rates, measured switching and inventory costs, introduce capacity constraints to turn the ranking into an optimisation, and pilot the top five moves to measure real delivery-time change.

VII. CONCLUSION

Shipping each product from the nearest capable factory, rather than a single fixed factory, would cut lead time by up to 39% on the busiest routes. Across 1,060 simulated scenarios, 42 routes are both faster and cheaper to serve from another factory, but only 10 repay a \$1,500 switch within a year; these save \$24,599 annually and coincide with the long-haul group found by clustering. The deployed application turns these results into a tool for testing a move, seeing its trade-offs and checking its risks before acting.

REFERENCES

- [1] S. Chopra and P. Meindl, *Supply Chain Management: Strategy, Planning, and Operation*, 6th ed. Boston, MA, USA: Pearson, 2016.

- [2] D. Simchi-Levi, P. Kaminsky, and E. Simchi-Levi, *Designing and Managing the Supply Chain: Concepts, Strategies, and Case Studies*, 3rd ed. New York, NY, USA: McGraw-Hill, 2008.
 - [3] S. H. Owen and M. S. Daskin, "Strategic facility location: A review," *Eur. J. Oper. Res.*, vol. 111, no. 3, pp. 423–447, 1998.
 - [4] M. T. Melo, S. Nickel, and F. Saldanha-da-Gama, "Facility location and supply chain management—A review," *Eur. J. Oper. Res.*, vol. 196, no. 2, pp. 401–412, 2009.
 - [5] M. A. Waller and S. E. Fawcett, "Data science, predictive analytics, and big data: A revolution that will transform supply chain design and management," *J. Bus. Logist.*, vol. 34, no. 2, pp. 77–84, 2013.
 - [6] G. Wang, A. Gunasekaran, E. W. T. Ngai, and T. Papadopoulos, "Big data analytics in logistics and supply chain management: Certain investigations for research and applications," *Int. J. Prod. Econ.*, vol. 176, pp. 98–110, 2016.
-
- ²<https://github.com/amarasuchitra/nassau-candy-webapp>
- [7] R. Carboneau, K. Laframboise, and R. Vahidov, "Application of machine learning techniques for supply chain demand forecasting," *Eur. J. Oper. Res.*, vol. 184, no. 3, pp. 1140–1154, 2008.
 - [8] R. W. Sinnott, "Virtues of the haversine," *Sky and Telescope*, vol. 68, no. 2, p. 159, 1984.
 - [9] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.
 - [10] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
 - [11] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Ann. Statist.*, vol. 29, no. 5, pp. 1189–1232, 2001.
 - [12] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York, NY, USA: Springer, 2009.
 - [13] A. M. Law, *Simulation Modeling and Analysis*, 5th ed. New York, NY, USA: McGraw-Hill, 2015.